

Rahul Singh

Applied AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

Applied AI / ML Engineer with **4+ years** building production vehicle-telemetry models and an internal RAG recommender. Developed fuel analytics deployed across **150,000 vehicles** and a financial research agent with bounded tool execution, evidence checks, and replayable evaluation.

TECHNICAL SKILLS

Generative AI / LLMs: LLM workflows, RAG, LangGraph, tool calling, structured outputs, LoRA / parameter-efficient fine-tuning

ML / NLP: PyTorch, TensorFlow/Keras, Hugging Face Transformers / PEFT, scikit-learn, LightGBM, sentence-transformers

Retrieval: Embeddings, semantic search, pgvector, Maximal Marginal Relevance (MMR) reranking

Backend & Data: Python, SQL, FastAPI, REST APIs, PostgreSQL, RabbitMQ

MLOps & Cloud: Docker, AWS, MLflow, OpenTelemetry, GitHub Actions (CI/CD), Git

PROFESSIONAL EXPERIENCE

Intangles — Machine Learning Engineer

Apr 2022 – Present

Pune, India **Spec Tag Recommender | Internal vehicle-tag recommendations**

- Engineered a production **RAG recommender** proposing 4–5 vehicle tags per request; users retained at least one suggestion on **97% of approximately 10,500 first requests** over three months. End-to-end p95 latency, including Bedrock inference, stayed **under one second**.
- Implemented the recommendation workflow in **LangGraph** and served it through **FastAPI**; retrieved historical manually assigned vehicle tags using sentence-transformer embeddings, a **PyTorch** projection network, and **pgvector**, with confidence tiers and MMR reranking.
- Built **RabbitMQ** recommendation and embedding consumers; added JSON parsing, type coercion, and deterministic checks to validate model-generated tags before team review.

Fuel Analytics | Customer-facing detection and volume estimation

- Developed customer-facing fuel analytics across **150,000 vehicles** and approximately **530,000 refill events monthly**, extending DBSCAN/LOESS detection on OBD telemetry.
- Reduced monthly theft-alert volume by **15%**, from **7,500 to 6,375 alerts**, after applying moving-average smoothing to fuel-level signals; evaluated matched three-month periods and reviewed customer/account-manager feedback.
- Built refill-volume estimation with accuracy within **±5% per eligible event** versus customer-entered filling logs; excluded events with missing logs, bad sensor readings, outliers, or failed estimates.
- Built **sub-threshold detection** for small thefts and refills missed by existing threshold logic and sensor limitations.

EV Battery Coolant Temperature | Team-based inference development

- Prepared irregular OBD/CAN telemetry and built ambient-temperature features for an existing **TensorFlow/Keras LSTM** coolant-temperature inference pipeline, producing contextual and forecast health states.
- Added physical-bound checks, output clipping, and deterministic fallbacks for missing inputs or degraded inference.

AI / MACHINE LEARNING PROJECTS

FIN-AI | CLI Financial Research Agent

github.com/rahul2-byte/financial-analyst-system

- Built a CLI-first NSE/BSE research agent with deterministic lookup routes and bounded LLM tool execution across **nine schema-defined tools**; kept financial calculations in Python rather than LLM-generated arithmetic.
- Implemented **fail-closed report-publication checks** for missing citations, unsupported major claims, and numeric facts without verified data; return limited-evidence results when checks fail.
- Built a **150-case live/replay evaluation**; one live run recorded **143/150 terminal successes (95.3%)** and **45.7s p50 / 110.3s p95** end-to-end latency; terminal success measures execution, not answer accuracy.
- Added provider snapshots and transcripts to inspect tool decisions and replay failures without live provider calls.

LoRA Reproduction & Parameter-Efficient RoBERTa Fine-Tuning Study

github.com/rahul2-byte/lora-reproduction

- Implemented and tested custom **PyTorch** LoRA adapters for RoBERTa query/value projections, including frozen base weights, rank/alpha scaling, adapter dropout, paper-style initialization, safe module injection, and merged inference export.
- Designed a reproducible **27-run** GLUE study: **18 core runs** across MRPC/SST-2 and three seeds, one **PEFT** reference run, and eight rank, scaling, and module-placement ablations.
- On one **RTX 4050** benchmark, reduced trainable parameters from **124.6M to 887K (99.29%)**, task artifacts from **498.7 MB to 3.57 MB (99.28%)**, and PyTorch peak allocated GPU memory by **61.9–64.6%** versus full fine-tuning.
- Across three seeds, achieved **92.81% ± 0.46%** SST-2 validation accuracy versus **93.04% ± 0.24%** for full fine-tuning; documented MRPC accuracy of **69.28% ± 0.86% vs. 87.17% ± 0.51%** and validated PEFT forward parity.

EDUCATION

Diploma in Computer Science and Engineering

Government Polytechnic Kashipur, Uttarakhand, India

Jul 2018 – Sep 2021